Making & Fining
Cyder

Timothy Matlack of Lancaster, 1807–1808

Two papers, both in *Memoirs of the Philadelphia Society for Promoting Agriculture*, vol. I (Philadelphia: Jane Aitken, 1808).
pp. 109–118 — the pommice press, two letters to Richard Peters, read 10 March 1807.
pp. 268–273 — making and fining, letter to Dr. James Mease, read 8 March 1808.

> He almost didn't write any of it down. Fermentation, he says, is less understood than any other subject philosophy or chymistry have attended to; Pennsylvania is too variable for directions to hold; and he expected the blame when other men followed his practice and failed. His excuse for speaking anyway, at seventy: to reason against fixed prejudices is folly that ought not to be expected beyond the age of 70. He had been trying apples for cider more than forty-eight years.

Mill & press

1807 PAPER

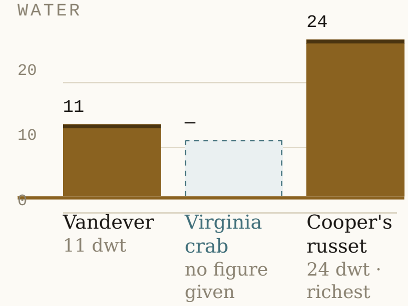
01 **Press immediately from the mill**
His flat contradiction of standard practice. The sooner pressed after grinding, the paler the cider, the more perfectly bright in the cask, the less lees in the bottle. A Society member footnoted this on the page as contrary to general opinion and worth experiments.

02 **Never time the pommace by the clock**
Judge by the heat of the air instead. The same interval that does nothing in one season starts an acid fermentation in a warmer one.

03 **Line the crib so the must runs fine**
Lattice bottom on the perforated floor, hair cloth or coarse bagging over it and on three sides of the crib.

Weigh the must

A PINT AGAINST A PINT OF RAIN WATER



The same 11-to-24 spread appears in both papers, in pennyweights heavier than rain water. The house apple is next after the russet. Dry seasons give a heavier must, wet seasons lighter.

He wanted farmers using Dicas's hydrometer and said they'd hardly credit the difference until they had one. A Society footnote supplies the name he'd forgotten: saccharometer.

Ferment

BOTH PAPERS · AUTUMN, FROM THE PRESS

01 **Keep it cool — his proof is from beer**
One malt wort split two ways. At 65° it came spontaneously fine and perfectly bright, soft, free of bitterness. At 76° it fixed a cloud no art could remove, decomposed enough for the hop resin to offend, and grew bitter as aloes.
65° bright · 76° ruined

02 **Skim the pulp as it rises**
Warmth is fermentation's first sensible effect; it expands the air in the pulp and floats it up. Take it off then. Its return drives the ferment harder, and pulp left in decomposes the cider into the exact bitter of an apple leaf.

03 **Watch the weather, not the calendar**
Attend to every change, above all cool to warmer. He refuses to give precise directions: the climate is too variable, and the best cellars run too warm in cider season.

04 **Restrain a runaway with spirits of wine**
Rectified spirits, burnt taste killed by powdered orris root. Lay it on the surface with the cask full to the bung hole.
1 oz orris : 1 pint · max 1 spoonful per hogshead

The anomaly that broke his own rule

The rule is heavier must, stronger cider. The Virginia crab breaks it — light must, cider equal to the best. Matlack's explanation isn't sugar at all: the crab's must has far less tendency to run away in fermentation than common apples of equal weight, so it needs less judgement to restrain. He calls that a decided superiority over every other apple, and leaves the cause for others to guess at.

FINING, SEALING & WARNINGS OVERLEAF →

Fine, seal & keep

Fine

BEFORE THE LAST OF MARCH

01 **Time it to a dead ferment**
Not the least degree of fermentation may be going on. That means finishing before the spring ferment begins — or waiting until it stops, by which point the cider has usually taken up enough acidity not to be worth fining at all.

02 **Make the fining**
Common staple isinglass, perhaps the safest fining. Pound it and break it into threads. Dissolve it in the cider to be fined, beating that cider well several times a day, then strain through a flannel bag.
5 staples : 2 hogsheads · beat 2–3 days

03 **Rack the cider onto the fining**
Fining into the empty cask first, then draw the cider off onto it. Three things at once: leaves much of the sediment behind, checks insensible fermentation, mixes cider and fining intimately. Fill quite full, then spirits of wine on the surface.

04 **Draw off the moment it goes bright**
Fine and bright in six or eight days. Draw off immediately and either bung close or bottle. Under-fermented cider will break your bottles.
6–8 days

Seal

THE GIMLET-HOLE TRICK

01 **Vent the bung while the cement is still warm**
Casks are bunged close and waxed over to keep air entirely out. Drive the bung carefully home, then bore a gimlet hole beside the bung hole and leave it open while you cover the bung with cement. Without the vent, the cement's warmth expands the air below, throws up a blister through the air hole and forever disappoints the attempt to close it. Once the cement has cooled and hardened, drive a square white oak plug into the gimlet hole.

What he tells you not to do

DON'T	DON'T	DON'T	INSTEAD
Expect it to keep like Madeira The best Madeira won't hold on less than eight gallons of brandy in the hundred, twelve more commonly. Unaided apple juice cannot be expected to stand through our summer's heat. Too much is expected of cider.	**Fortify and hope it hides** Brandy blends into wine over years until the palate stops finding it. Cider does the opposite — it exposes even the smallest mixture of the best brandy, or any known spirits, more readily than any other liquor.	**Boil the must down** Concentrating by long boiling makes added yeast absolutely necessary to ferment it back to a drinkable degree — and he never met one his stomach didn't complain of, even after a moderate draught.	**Concentrate by frost** On cider already duly fermented, in our coldest weather: a delicate bottled liquor that will stand for years and improve by time.

His press — 1807 paper, pp. 109–118

Built for currant, blackberry, grape and quince wine, sized up for perry, then fitted to cider "in a way far indeed beyond my first intention." A hundred-pound weight geared five by five presses 2,500 lb downward; because the uprights are fastened to the side planks, each lever's toe bears the crib upward with the same power the heel presses down — so the pommace actually takes 5,000 lb.

`100 lb → 5,000 lb on the pommace` `two levers → 10,000 lb` `four → "and so infinitely"` `crib 21 × 20 × 20 in` `presses like 37 bushels` `whole press 13 ft 7 in` `2 days' carpentry` `$2.50 of plank` `a chain, a pin, 100 4d nails` `packs in a box 20 in sq × 8 ft`

Any farmer who can handle a saw, an axe and an augur can make the whole thing — a strong withe will serve for the chain, tough hickory for the iron pin. He built both the working press and the Society's demonstration model himself, knowing it would take longer to get a mechanic to do it. Engraved plate at p. 115; keyed description pp. 116–118.

Sources. Both papers in *Memoirs of the Philadelphia Society for Promoting Agriculture*, vol. I (Philadelphia: printed by Jane Aitken, 1808) — "Account of a new Pummice Press, with some remarks upon Cyder making," pp. 109–118, and "Observations on making and fining Cyder, and on Peach Trees," pp. 268–273, which runs on to p. 279 on peach trees. Full scan at archive.org/details/memoirsofphilade1808phil. Every step and figure here is Matlack's own; nothing has been supplied from other period sources. "Bright," he notes, is a brewer's word — not merely fine, but the perfect transparency in which liquors are tasted in their purity.